

Munkres 2

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1 Introduction

16.9: Let L be a line in the plane $\mathbb{R}^2$. Find all topologies that L inherits from $\mathbb{R}_l \times \mathbb{R}$ and $\mathbb{R}_l^2$.

The ordered square topology I_o^2 is the order topology on the closed unit square in dictionary order.

If S is open in $X \times Y$, is $\pi_1(S)$ open in X ? If S is closed, is $\pi_1(S)$ closed? In the first case, $\pi_1(S)$ is open, but in the second, $\pi_1(S)$ need not be closed. If X is finite, is it true if S is closed, then $\pi_1(S)$ is also closed?

How to motivate proofs that a set is not metrizable?

Let $f_n : X \rightarrow \mathbb{R}$ be a sequence of functions such that $|f_n(x)| \leq M_n$ for all $x \in X$ and $\sum_{i=1}^{\infty} M_i$ converges. Define $s_n(x) = \sum_{i=1}^n f_i(x)$. Then (s_n) converges uniformly to a function s .

Supplementary 7.